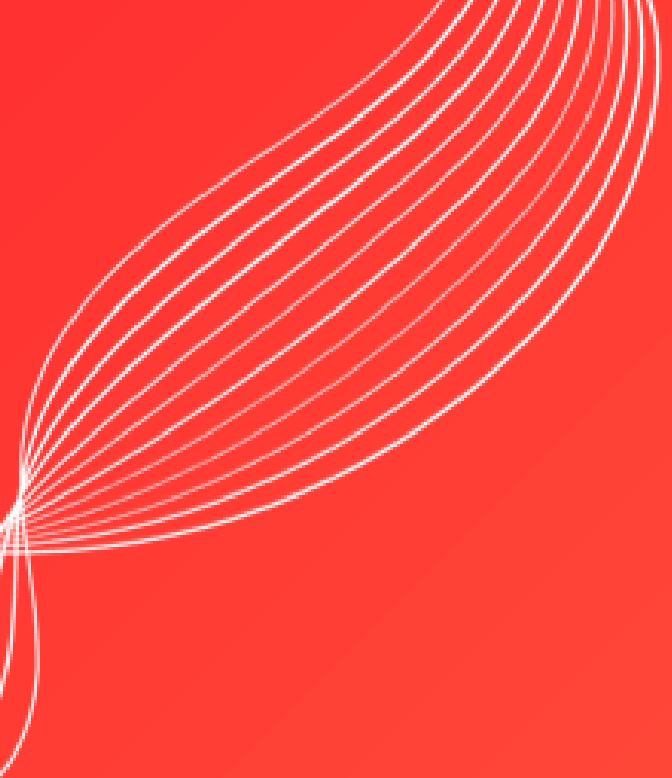




LuxeMod

Insights & Recommendations

GCOM 7140 — Customer Analytics

By: Enmanuel Baez, Tenzin Lodoe, Jad Shaheen, Luke Sherrill, and Kalsang Tseten



Introduction

Company Overview:

- Digital-first furniture retailer for young professionals and families
- Focus on modern/modular designs for urban living spaces
 - Emphasis on pieces that serve multiple functions
- Product lines cover main interior rooms (bedroom, bathroom, etc.)

Business Problems:

- High cart abandonment rates
- Challenges with customer retention
- Inconsistent customer satisfaction scores
- Need to optimize marketing spending

Objective:

- Analyze customer behavior, product category trends, and digital engagement metrics to uncover key insights
- Create predictive models to forecast cart abandonment rates and satisfaction scores
- Form two recommendations to enhance customer experience, reduce cart abandonment, and maximize

Agenda



- 1.Dataset Overview**
- 2.Data Cleaning**
- 3.Exploratory Data Analysis**
- 4.Predictive Modeling**
- 5.Roadblocks & Resolutions**
- 6.Strategic
Recommendations**
- 7.Q & A**

Dataset Overview

Variable	Description	Variable	Description
Satisfaction Score	Rating value for customer satisfaction	Age	Customer's age in years
CLV	2-year projection in USD of a customer's value	Purchase Frequency	Number of purchases made to date by customer
Cart Abandonment	Customer proportion of shopping carts abandoned	Income	Customer annual household income in USD
Months Active	Number of months as an active customer	Average Spending	Customer's average amount spent per order in USD

Data Cleaning

1. Converted Features to Appropriate Data Types

2. Identifying Features with NA Values

3. Regression Model to Impute NAs

4. Logistic Regression to Impute NAs

Converting Features to Appropriate Data Types

- 5 Features
- Converted from Strings to Logicals

```
luxemod <- luxemod |>
mutate(
  homeowner = ifelse(homeowner %in% c("Y", "Yes"), TRUE,
    ifelse(homeowner %in% c("N", "No"), FALSE, NA)) |> as.logical(),

  living_room = ifelse(living_room %in% c("Yes", "1"), TRUE,
    ifelse(living_room %in% c("No", "0"), FALSE, NA)) |> as.logical(),

  bedroom = ifelse(bedroom %in% c("Yes", "1"), TRUE,
    ifelse(bedroom %in% c("No", "0"), FALSE, NA)) |> as.logical(),

  dining_room = ifelse(dining_room %in% c("Yes", "1"), TRUE,
    ifelse(dining_room %in% c("No", "0"), FALSE, NA)) |> as.logical(),

  office = ifelse(office %in% c("Yes", "1"), TRUE,
    ifelse(office %in% c("No", "0"), FALSE, NA)) |> as.logical()
)
```

Identifying Features with NA Values

- The table displays the number of missing values for each column, revealing patterns in data gaps.
- The "Age", "Income", and "Homeowner" features have 50, 30, and 307 missing values respectively.

customer_id	age	income	homeowner	household_size
0	50	30	307	0
purchase_frequency	avg_order_value	total_spent	return_rate	sessions_per_month
0	0	0	0	0
avg_session_duration	cart_abandonment_rate	living_room	bedroom	dining_room
0	0	0	0	0
office	satisfaction_score	months_active	clv	
0	0	0	0	

Regression Model to Impute NAs

- impute_missing_values Function Created
- Predict and impute Age and Income NA values.

```
impute_missing_values <- function(df, target_column, include_homeowner = 1) {  
  # Having an option to run without homeowner for the data points that are missing both  
  # homeowner and either age / income  
  if (include_homeowner == 0) {  
    df_filtered <- df %>% select(-homeowner)  
  
    train_data <- df_filtered %>% filter(!is.na(!!sym(target_column)))  
    test_data <- df_filtered %>% filter(is.na(!!sym(target_column))) %>% select(-all_of(  
      target_column))  
  }  
  else{  
    train_data <- df %>% filter(!is.na(!!sym(target_column)))  
    test_data <- df %>% filter(is.na(!!sym(target_column))) %>% select(-all_of(target_column))  
  }  
  
  formula <- as.formula(paste(target_column, "~ ."))  
  model <- lm(formula, data = train_data)  
  predicted_values <- predict(model, newdata = test_data)  
  
  # Fill missing values in the original dataset  
  df[is.na(df[[target_column]]), target_column] <- predicted_values  
  
  return(df)  
}
```


Logistic Regression to Impute NAs

- A logistic regression model was trained to predict Homeowner status using all the features in the dataset.
- Results
 - > 0.5 -> True
 - < 0.5 -> False

```
{r}
luxemod <- luxemod |> mutate(homeowner = as.factor(homeowner))
train_data <- luxemod |> filter(!is.na(homeowner))
predict_data <- luxemod |> filter(is.na(homeowner))
```

```
{r}
logit_model <- glm(homeowner ~ age + income + household_size:cart_abandonment_rate +
satisfaction_score + months_active + clv, data = train_data, family = binomial)
```

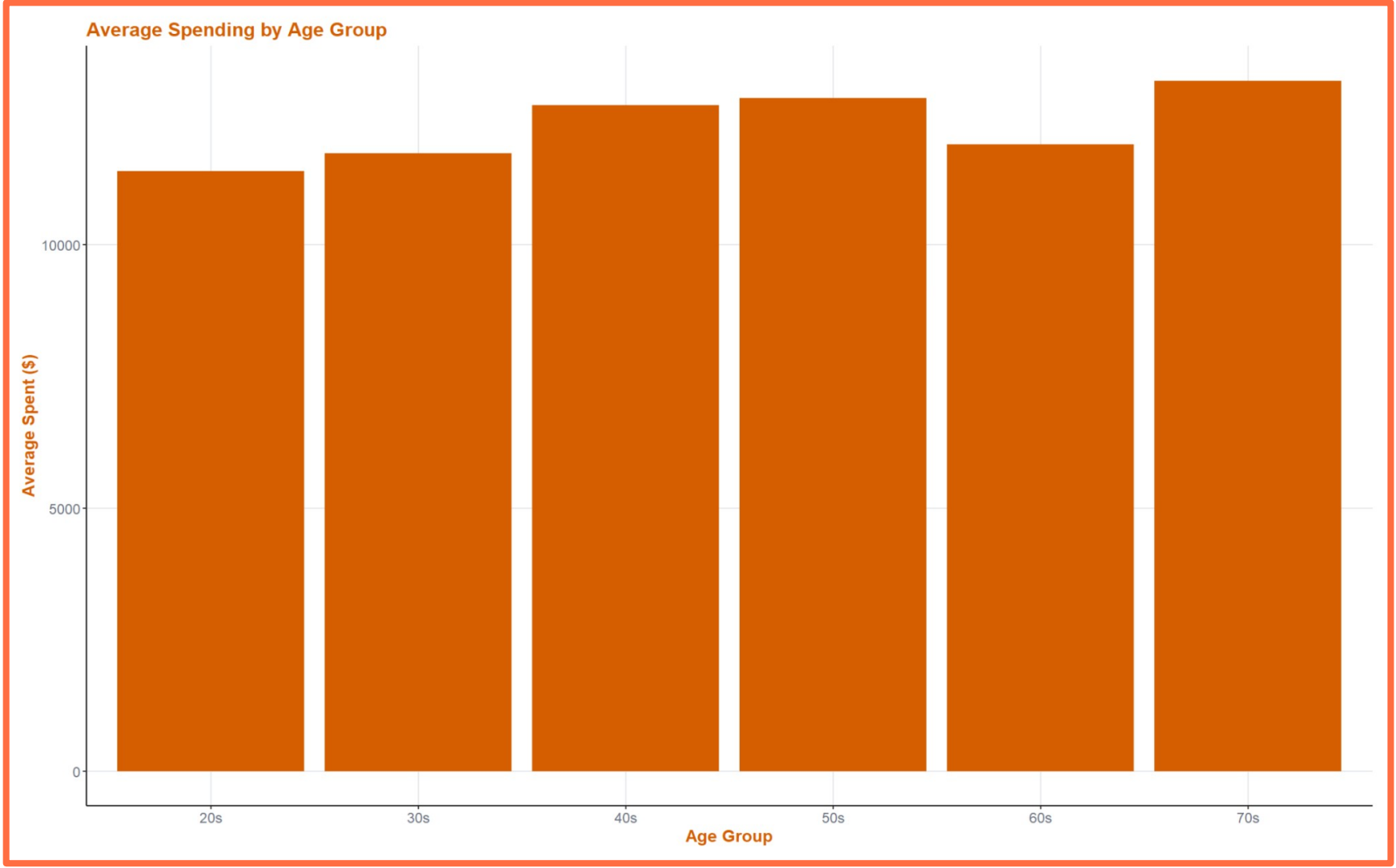
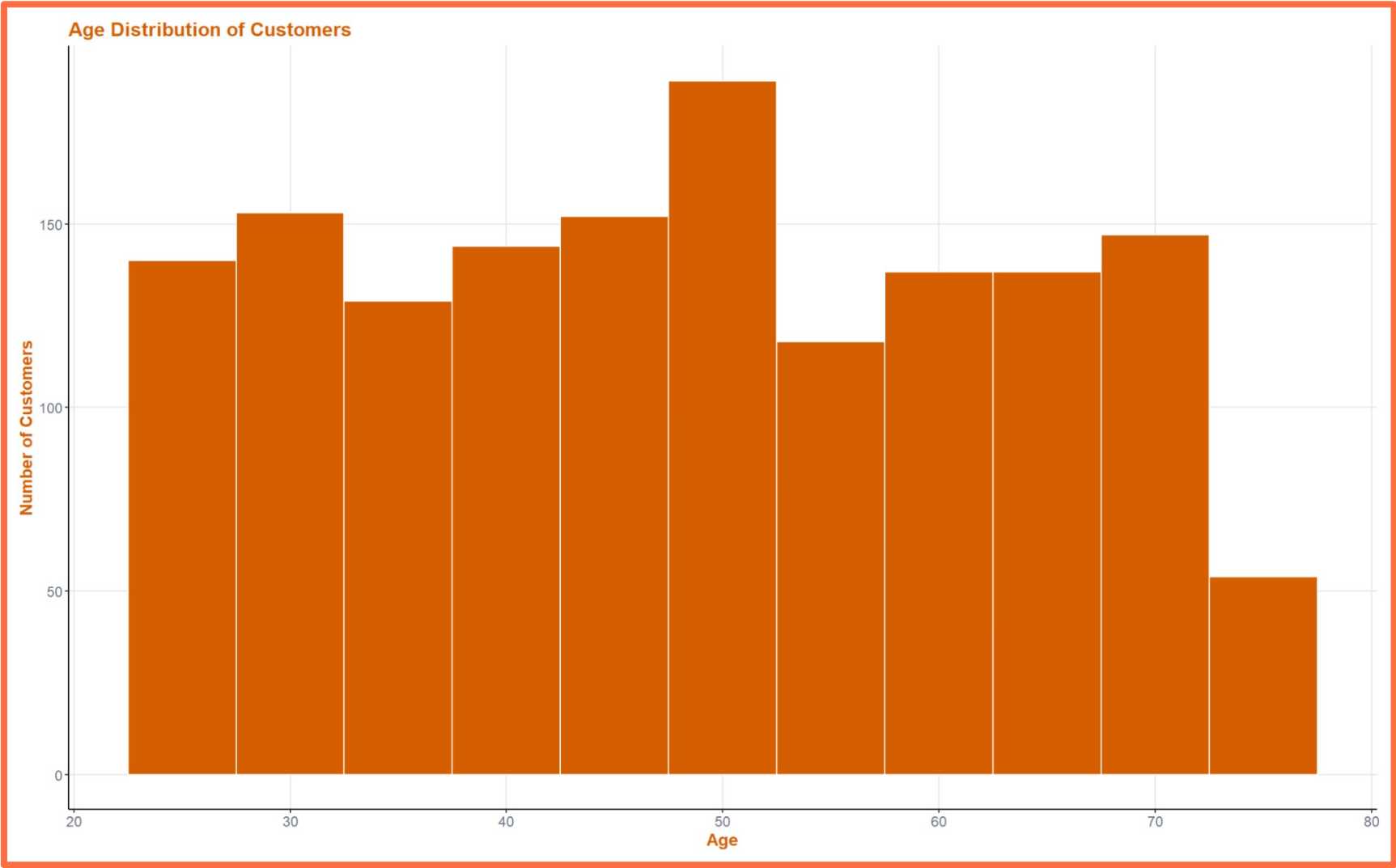
```
{r}
predict_data$homeowner_pred <- predict(logit_model, predict_data, type = "response")
predict_data$homeowner <- ifelse(predict_data$homeowner_pred > 0.5, TRUE, FALSE)
predict_data <- predict_data |> select(-homeowner_pred)
```

```
luxemod <- luxemod |>
  mutate(homeowner = as.logical(homeowner)) |>
  anti_join(predict_data, by = join_by(customer_id) ) |>
  union_all(predict_data)
```

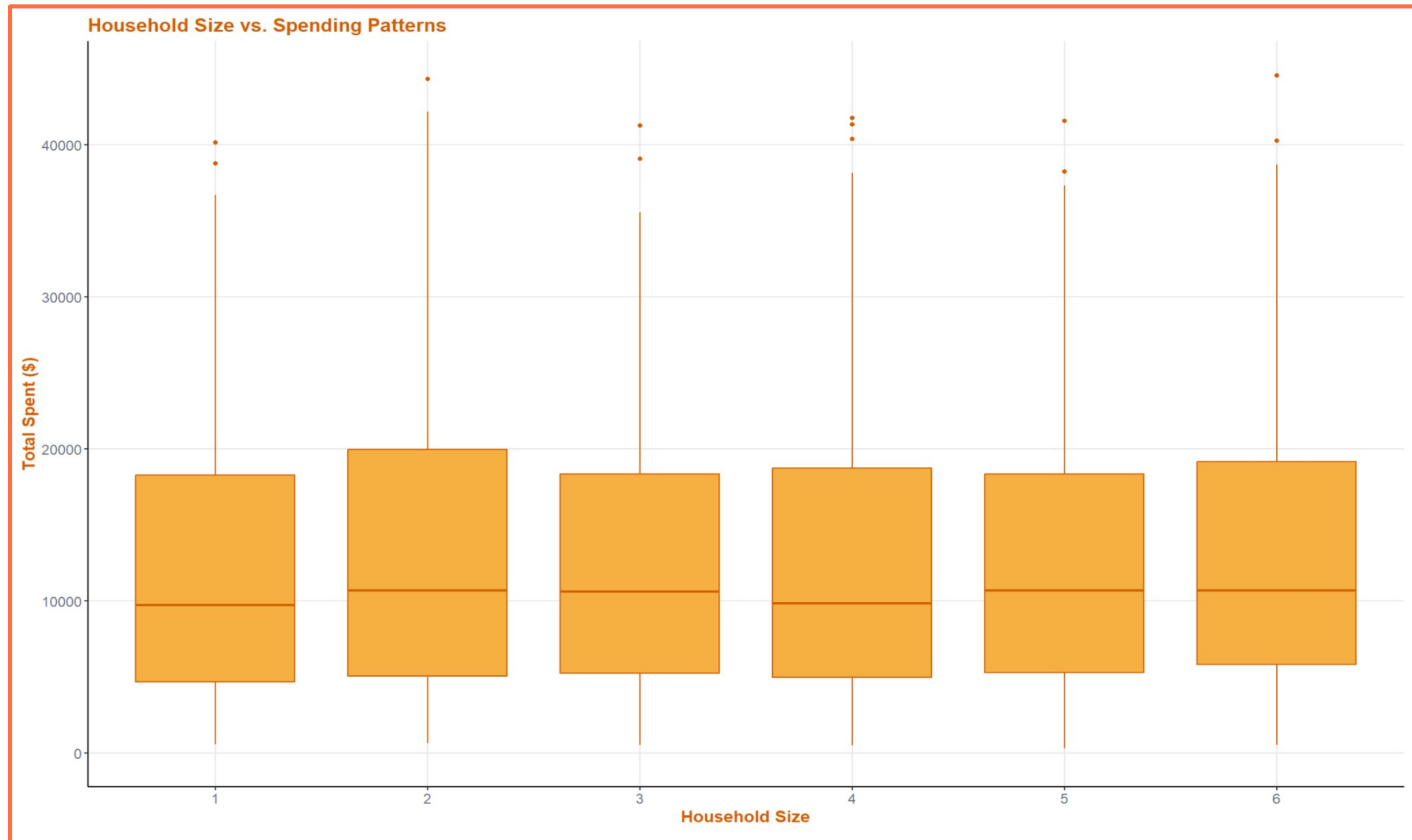

Exploratory Data Analysis (EDA)

- 1.Customer Segments**
- 2.Products and Services**
- 3.Customer Lifetime Value (CLV)**
- 4.Cart Abandonment Rate**

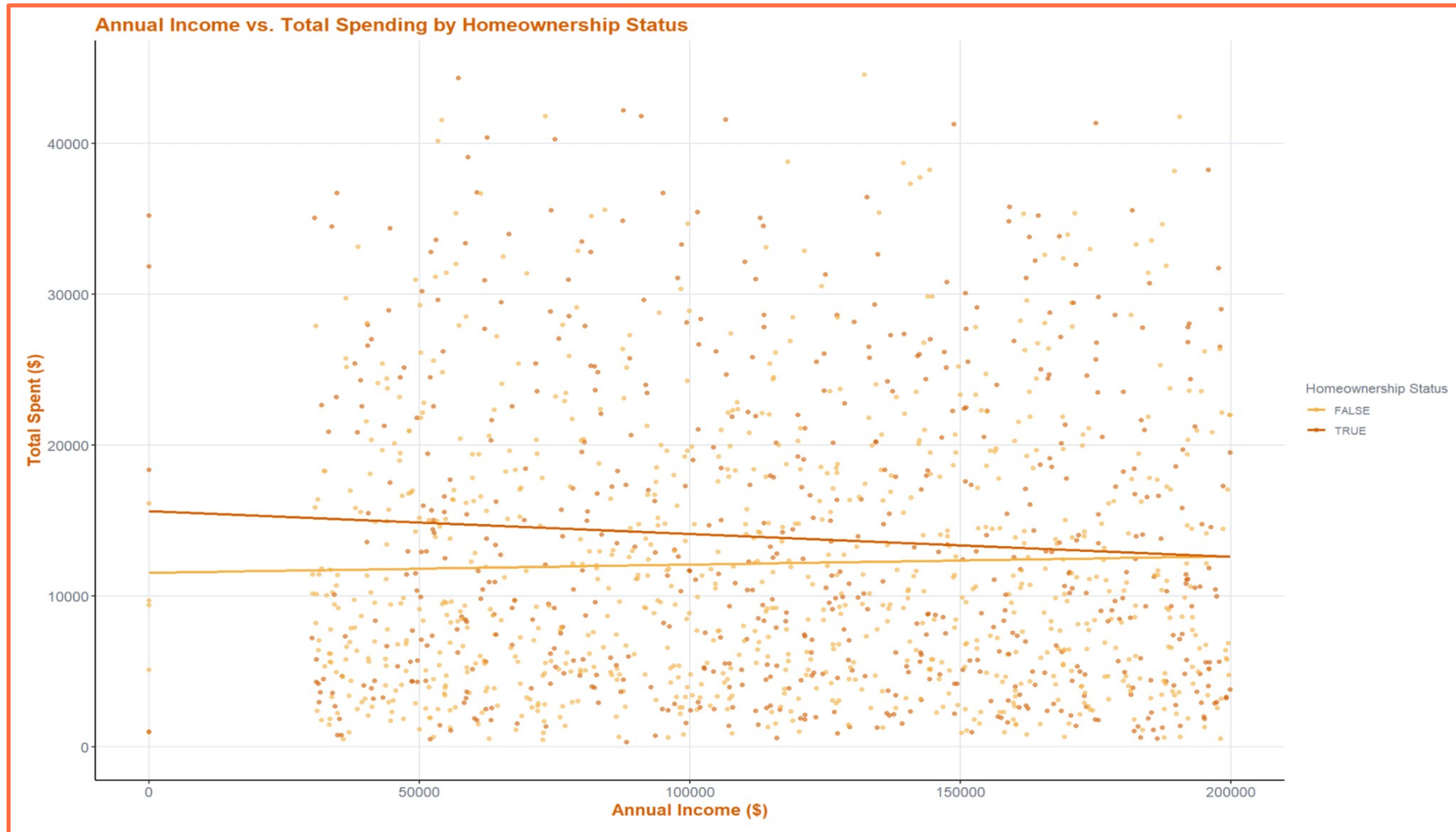
Customer Segments



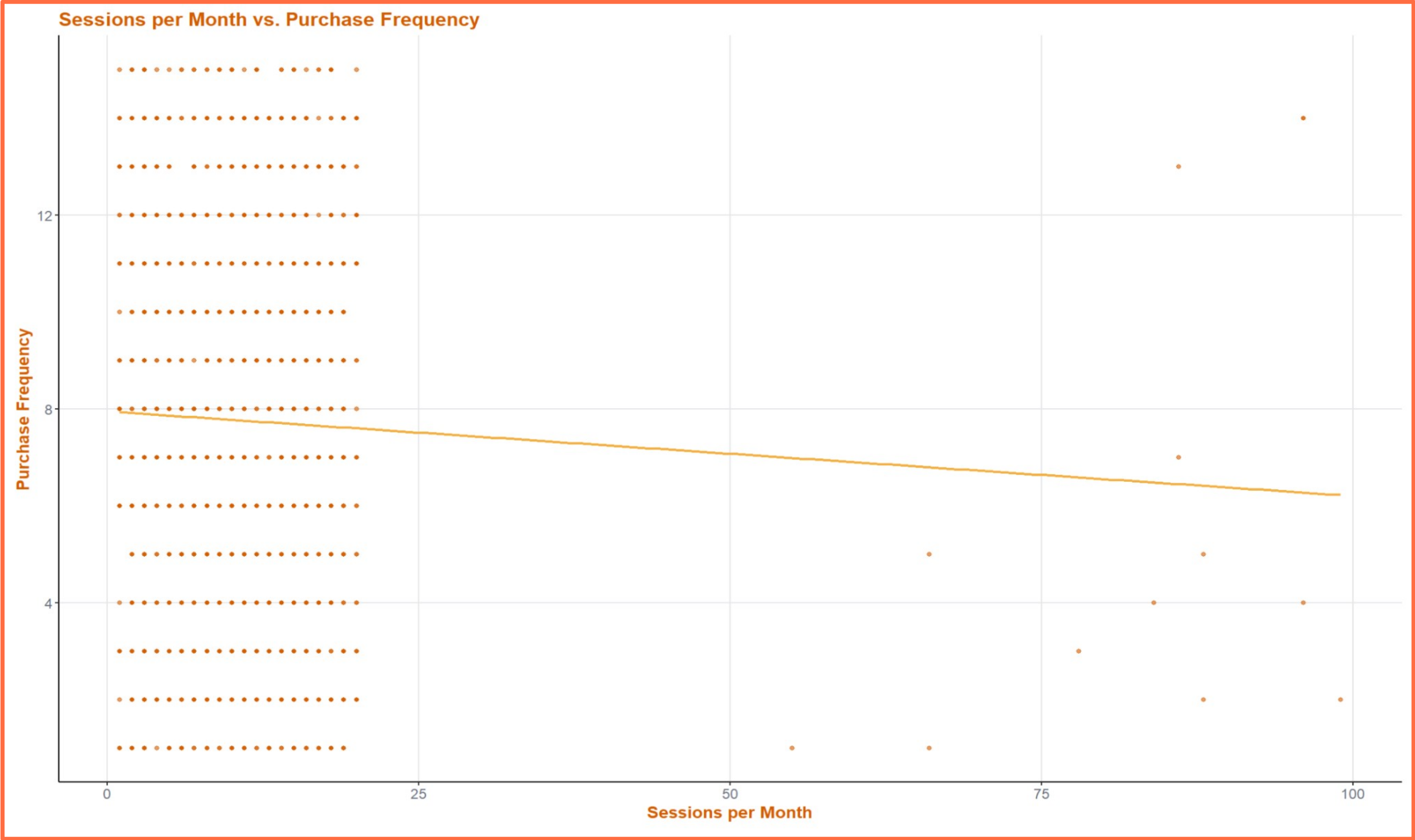
Customer Segments



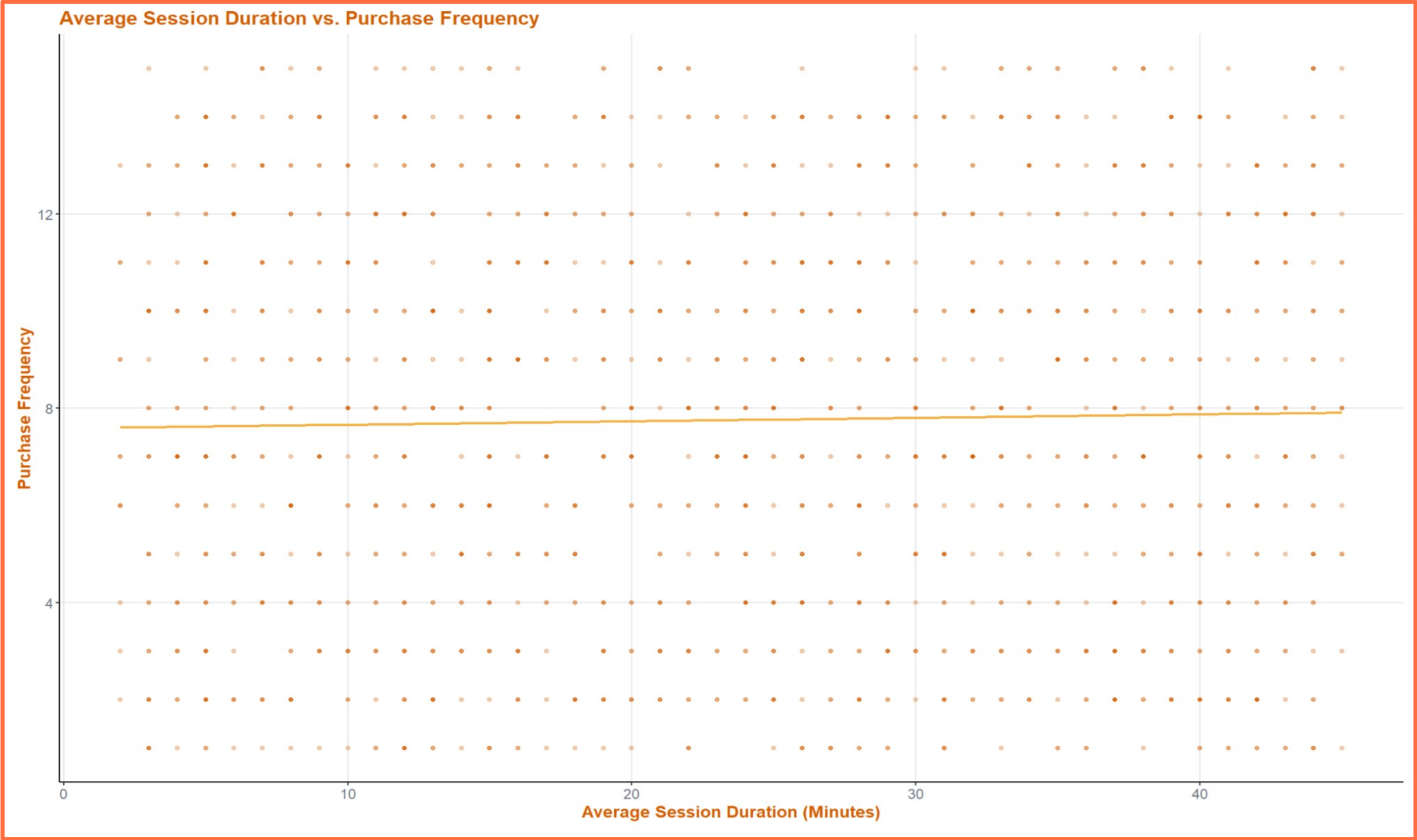
Customer Segments



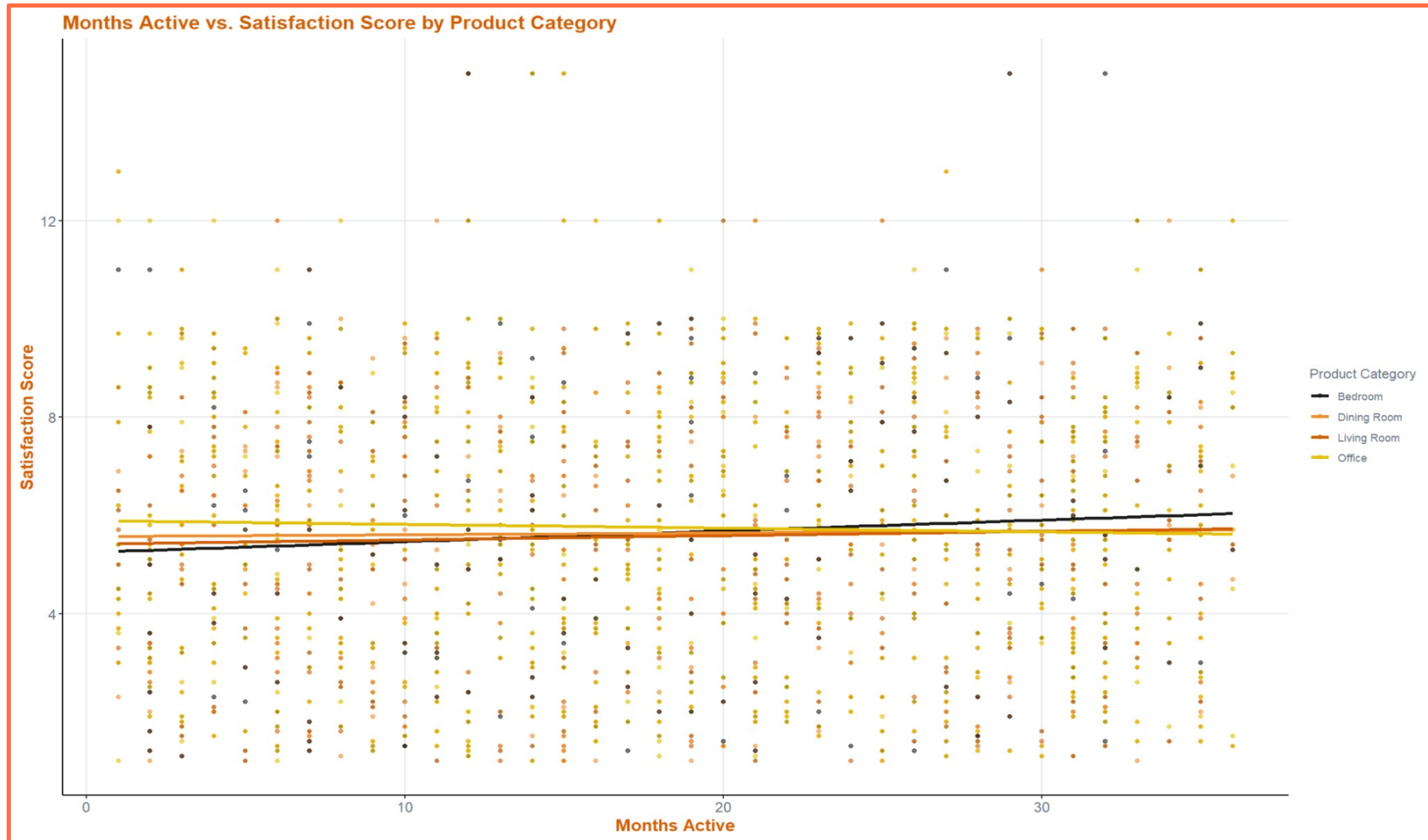
Products & Services



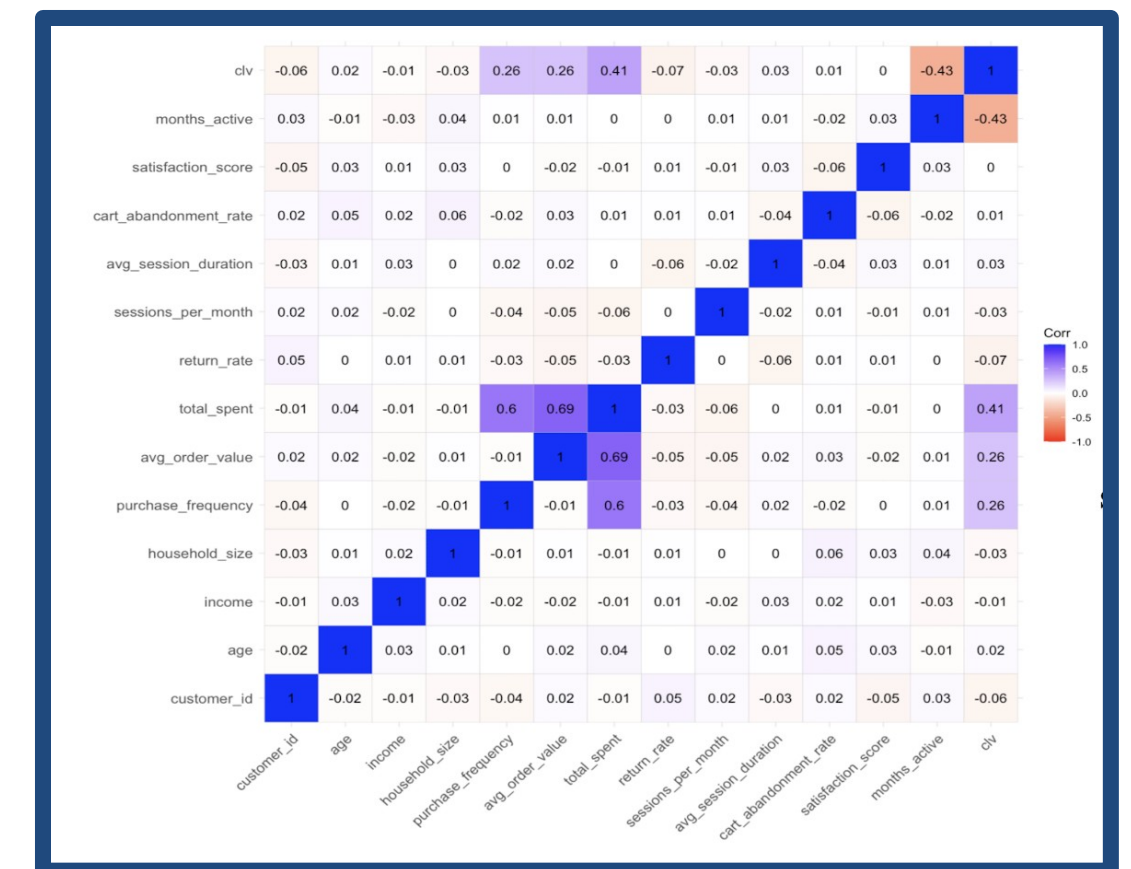
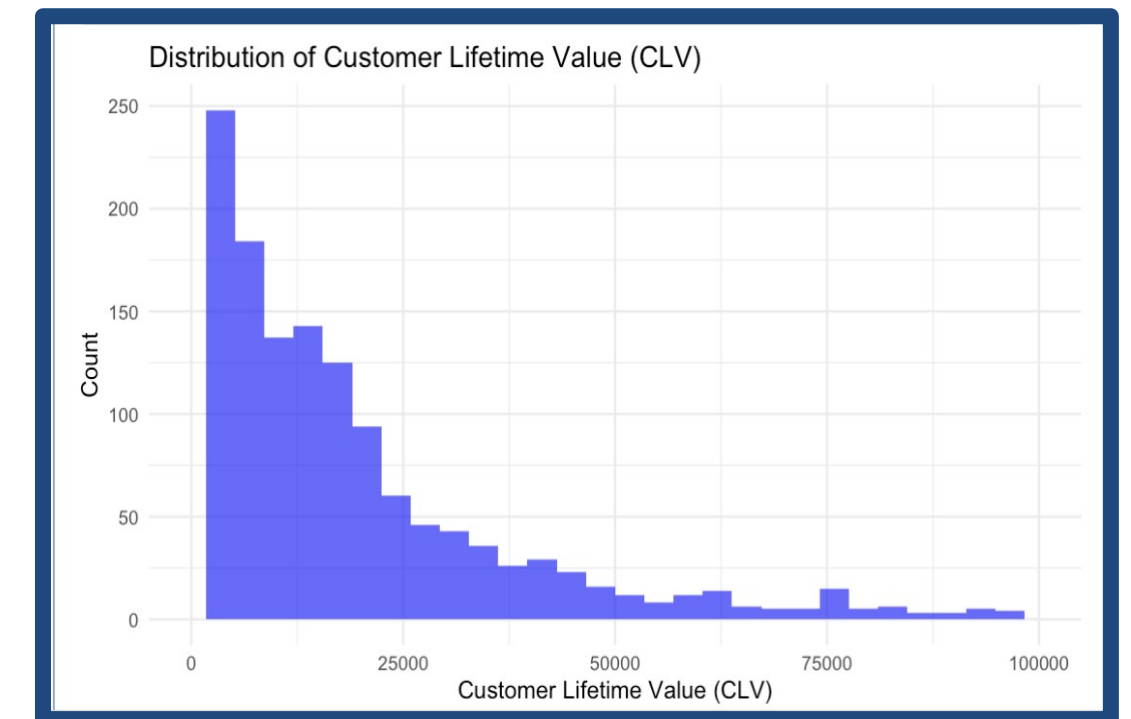
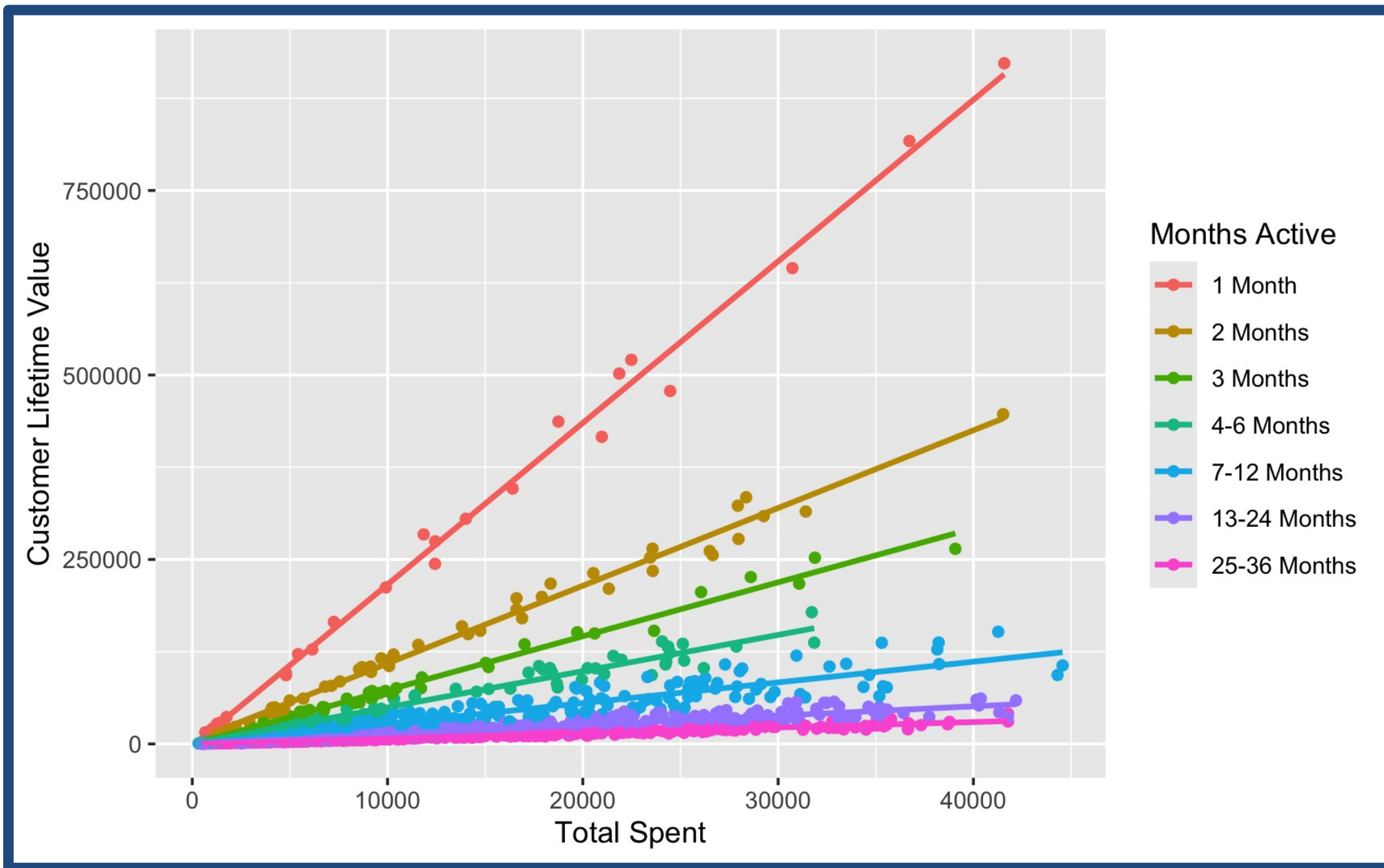
Products & Services



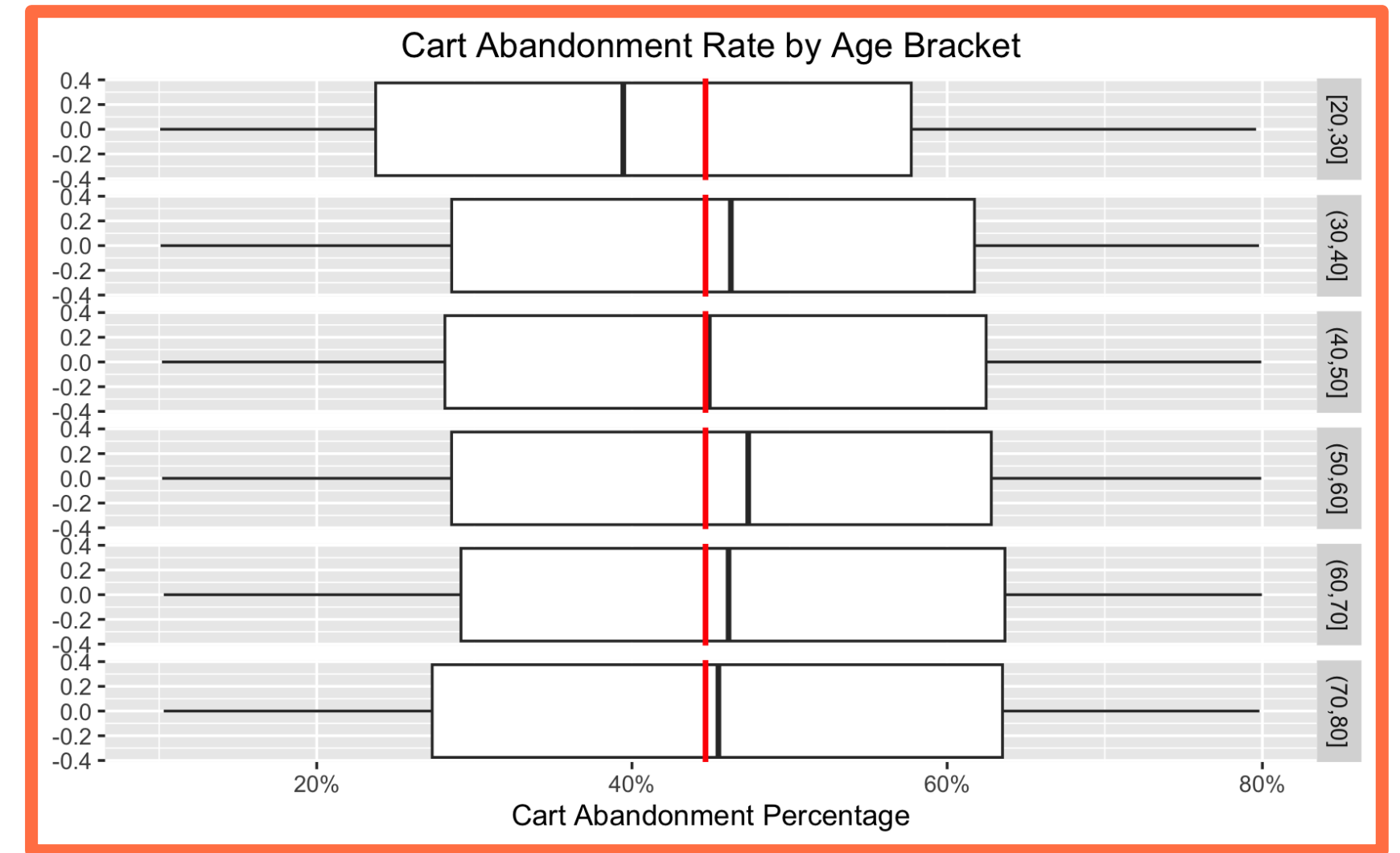
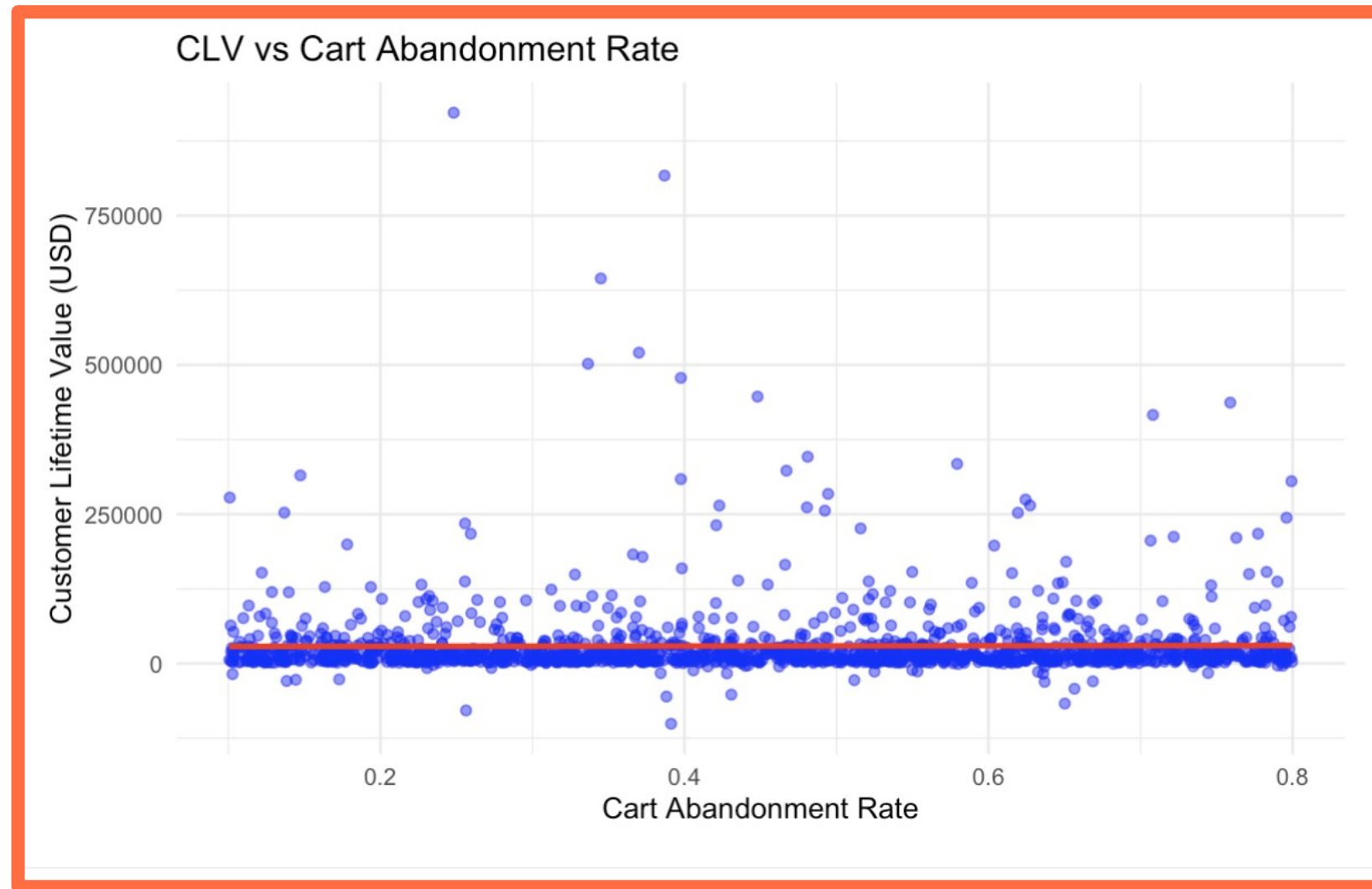
Products & Services



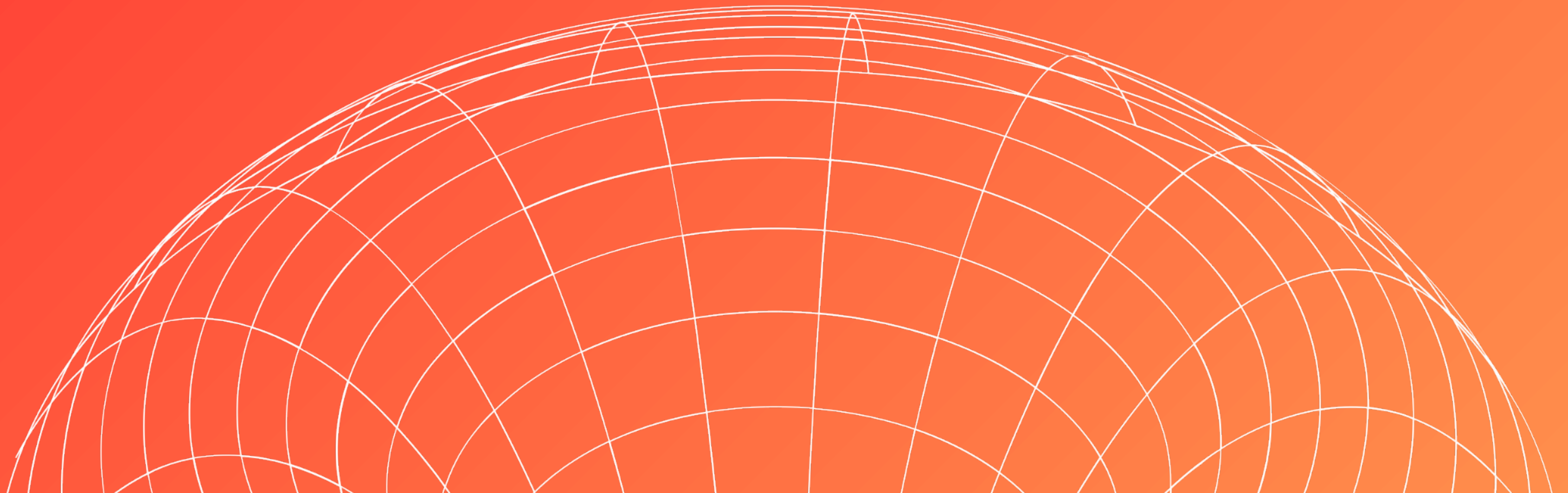
Customer Lifetime Value (CLV)



Cart Abandonment Rate



Predictive Modeling



Predictive Modeling

Decision Tree

Identify Cart Abandonment Driving Factors

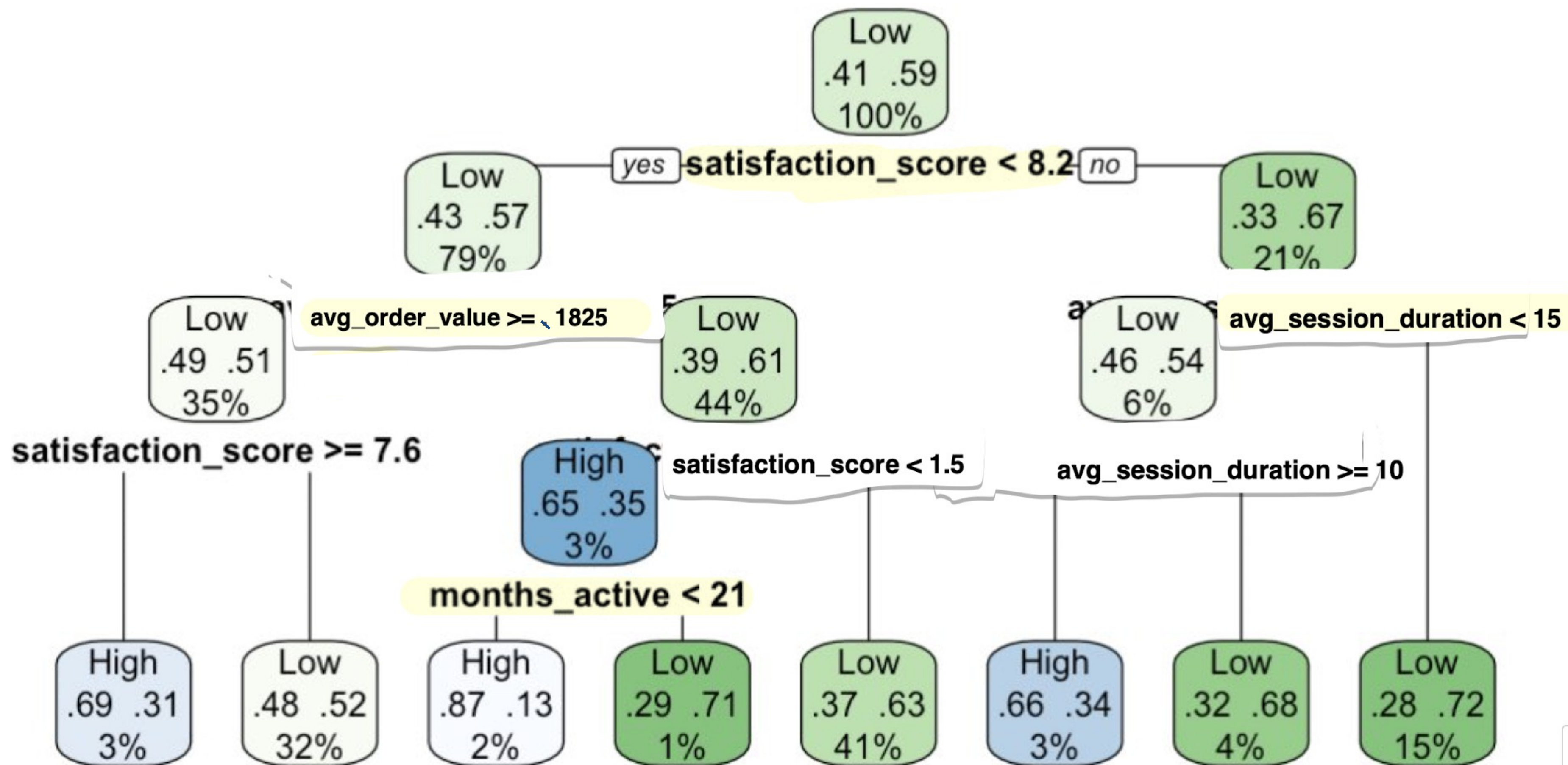
- By training a decision tree, we can determine which customer behaviors (e.g., session duration, satisfaction score, homeowner status) most influence whether a user abandons their cart.

Make Data Decision Business Decisions

- Users with higher average order values and longer session durations tend to have a lower probability of abandoning their carts.



Cart Abandonment Prediction Decision Tree



Predictive Modeling Summary

Customer Satisfaction

- Customers with lower satisfaction scores (< 8.2).
- Shorter session durations (< 15 minutes) are more likely to abandon their carts.

Cart Abandonment Rate

- High cart abandonment occurs when customers have low satisfaction scores (< 7.6), short browsing times, or low engagement.
- Low cart abandonment is seen among longer active users (> 21 months) and those with higher satisfaction scores.

Strategies

- Retention-focused strategies can target long-term customers with higher satisfaction to reduce abandonment rates.

Roadblocks & Resolutions

Stat. Significance - Correlation Analysis - Feature Engineering	Imputing NA Values - Creating models to impute missing values	Finding EDA Trends - Many EDA illustrated relationships with very weak associations - Made forming suggestion more difficult
Homeowner Variable - Difficulty converting Categorical Data into a Binary for the Decision Tree	Platform Compt. - Code must be cleaned to be compatible across all devices - Ensures reproducibility and cross-platform engagement	Insufficient Data - No data on marketing and customer acquisition costs even though it was one of the biggest challenge



Strategic Recommendations

Recommendation 1

Strategies to Attract & Convert High-Value Customers

● **Prioritize new customers**

- Launch a Referral Program
 - Reward existing customers for bringing in new buyers.
 - Offer discounts, store credits, or exclusive perks for successful referrals.
- First-Purchase Discounts and Limited time bundles
- Targeted follow-up campaigns

● **Don't forget about your old customers**

- 3+ months = \$26.3M vs 0-3months = \$17.5M
- Loyalty program
- Exceptional products and services helps build and maintain trust

Recommendation 2

How to target customers and optimize marketing costs?

- **Don't target demographics** - though it may seem intuitive

- **Do target shopping behaviors**

 - Purchase Frequency

 - ~ \$3,919 increase in CLV per additional purchase

 - Average Order Value (AOV)

 - \$17 increase in CLV per \$1 spent per order

- **Conclusion:**

1. Focus on repeat purchases and higher order values.

2. Implement loyalty programs & bundling strategies to encourage larger transactions

```
Call:
lm(formula = clv ~ age + income + purchase_frequency + avg_order_value + return_rate + satisfaction_score, data = luxemod)

Residuals:
    Min       1Q   Median       3Q      Max
-98449 -22685 -11249   4300  845164

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -2.820e+04  8.106e+03  -3.479 0.000518 ***
age           7.938e+01  9.862e+01   0.805 0.421003
income       -9.109e-03  2.928e-02  -0.311 0.755787
purchase_frequency  3.919e+03  3.610e+02  10.856 < 2e-16 ***
avg_order_value  1.727e+01  1.578e+00  10.944 < 2e-16 ***
return_rate   -3.963e+04  2.526e+04  -1.569 0.116974
satisfaction_score  1.771e+02  5.259e+02   0.337 0.736345
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 56820 on 1493 degrees of freedom
Multiple R-squared:  0.1409,    Adjusted R-squared:  0.1374 
F-statistic: 40.81 on 6 and 1493 DF,  p-value: < 2.2e-16
```

Next Steps

- Gather additional data to refine customer acquisition cost (CAC) estimates and marketing spend tracking.
- Analyze and optimize marketing budget allocation by identifying high-ROI customer segments and reducing spend on low-value audiences.
- Improve tracking of acquisition sources to determine which marketing channels drive the highest CLV.
- Refine retention strategies based on CLV insights, focusing on repeat purchases and loyalty incentives for top customers.



Thank You!

Any Questions?